



Capstone Progress Report 1

Stakeholder Names and Roles

Stakeholder	Role
<i>Eric Rodriguez</i>	<i>Team member</i>
<i>Hithesh Yedlapati</i>	<i>Team member</i>
<i>Elena Tsvetkova</i>	<i>Team member</i>
<i>Mohammad Farooq</i>	<i>Team member</i>
<i>Richard Carlson</i>	<i>Sponsor – Data Contact</i>
<i>James Garrison</i>	<i>Sponsor</i>
<i>William Basener</i>	<i>Mentor</i>

Project Title: Analysis of Sailplane Flight Logs to Evaluate In-Flight Engine Start Procedures and Pilot's Operational Behavior

Abstract

This project analyzes thousands of IGC flight log files to enhance sailplane safety by evaluating in-flight engine start procedures and pilot adherence to manufacturer guidelines. In-flight engine starts are critical for pilots returning to base, yet little statistical analysis exists on their compliance. By identifying patterns in real-world flight data, detecting anomalies, and assessing pilot decision-making, this project aims to improve operational best practices and safety protocols. The insights gained will help pilots optimize engine start procedures and support aviation safety organizations in developing data-driven recommendations for safer sailplane operations.

Outline of the Project

Sailplane pilots must perform in-flight engine starts to return to base, but failed starts can pose safety risks and may result in off-field landings. This project will analyze IGC flight logs to assess pilot adherence to manufacturer-recommended engine start procedures, identify anomalies, and improve best practices for safe operations. By understanding pilot behavior and identifying trends in successful and failed starts, we can improve training, reduce risk, and enhance safety protocols.

The primary stakeholders are aviation safety organizations, glider manufacturers, and sailplane pilots. Pilots will benefit from clearer best practices, while safety organizations and manufacturers can use these findings to refine recommendations and improve aircraft design.

An important assumption is that the IGC flight log data accurately captures key variables such as altitude, speed, and engine usage. Additionally, wind vector estimation through circling events is assumed to be reasonably accurate. This project is in scope for analyzing IGC logs, identifying trends, and classifying engine start attempts. However, it does not include real-time monitoring solutions.

Success Criteria

SC1	<i>Determine whether the community is adhering to the manufacturer's guidelines for in-flight engine operations</i>
SC2	<i>Establish a method to determine whether an in-flight engine start attempt was successful or failed</i>

Data Assumptions and Limitations

Right now, the main data limitation we are having is that the data does not factor in true wind speed, so we are having to calculate that, which is proving to be difficult. Wind speed and directors, also known as wind vectors were calculated based on equidistant pairs of points during circling events. We had to assume that the wind vector during this circling period is the same for the whole flight. We calculated the ground speed vector minus the wind vector to get instantaneous speeds during engine events. The error percentages using this approach is shown below. The reason we had to calculate speeds during engine events is to get a large enough sample size for determining false engine runs. Also, we are having to determine whether circling patterns from the gliders affected our overall speed and how it affects the question we are trying to answer. Also, each glider seems to have its own requirements which can make simplifying the question difficult. We are not having trouble accessing our data and Rivanna is so far providing us with enough resources to work with the data. There are only about 2000 flights out of 30,000 with engine runs. This makes it difficult to make a model that is high performing and statistically significant. Another limitation is that the glider names are all different in each IGC file. We had to do a lot of data cleaning with glider names to ensure that what is written in the IGC file is the same as in our standardized list of glider names. Some of these replacements may not be entirely accurate, and due to the nature of how these glider names were written, we had to go with what we believe the standardized version was. For example, "D2CT" is actually "DISCUS-2CT". We had to make over 100 replacements, and some of these replacements may not all be correct, but we did the best we could based on the information that was provided.

Summary of Data Processing, Data Aggregation

The target variable is determining if an in-flight engine start attempt succeeded or failed. This will be determined based on flight data patterns. The raw data comes in the form of thousands of IGC files showcasing flight information, such as timestamps, position coordinates, altitude, and speed. Potential predictor variables in this dataset are speed (both flight speed and ground speed), altitude, and geolocation. All of these can help us when we are looking to answer our questions. Our data was obtained from our sponsors who aggregated the IGC files into a memory drive for us to upload to Rivanna, UVA's cloud computer. Rivanna allowed us to simultaneously work on the data after which the data was aggregated based on glider type. We quickly realized each glider has its own specification and each has its own validation method. We started by grouping by glider type and did some data cleaning to get a sample for which we are currently testing on.

Data Visualizations

To analyze and interpret the flight data, we used **SeeYou Cloud**, which is a visualization tool designed specifically for sailplane/glider pilots. The platform allows us to upload an IGC file and will generate an interactive map displaying the flight's path, along with key metrics such as altitude, air speed, wind speed, engine noise level, etc.



By leveraging this tool, we can track a pilot's actions in real time and assess whether they are adhering to the recommended in-flight engine procedures. As shown in the image above, the flight path and altitude data provide insights during an engine run allowing us to identify trends and potential safety risks.

Summary of Modeling and Analysis

For flights with engine runs, this is the proportion of flights that started their engine above 1000ft. In other words, this is the number of flights whose altitude in feet was above 1000ft during the first time they started their engine for gliders whose launch type was Turbo and the second time they started their engine for gliders whose launch type was Self-Launching. We excluded pure gliders in our analysis.

```
Engine_Start_Above_1000ft
True      225
False     199
```

For 118 flights with TAS and engine runs, here are the error percentages for mean, max and min estimated TAS. Error percentages were calculated as: $((\text{calculated value} - \text{true value}) / \text{true value}) \times 100$. We identified a circling event for all 118 flights.

For 53 flights with Turbo Launch-Type:

Mean Error Percent Differences:

```
Min_%%_off  17.63
Max_%%_off  13.64
Avg_%%_off  15.17
```

Median Error Percent Differences:

```
Min_%%_off  12.98
Max_%%_off   9.20
Avg_%%_off  13.43
```

For 65 flights with Self-Launch Launch-Type:

Mean Error Percent Differences:

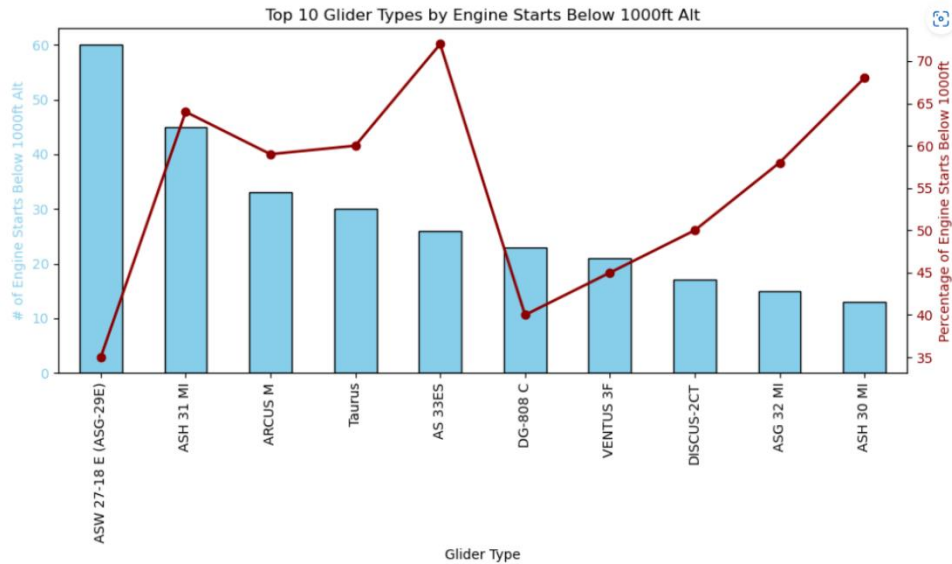
```
Min_%%_off  16.62
Max_%%_off  15.87
Avg_%%_off  13.89
```

Median Error Percent Differences:

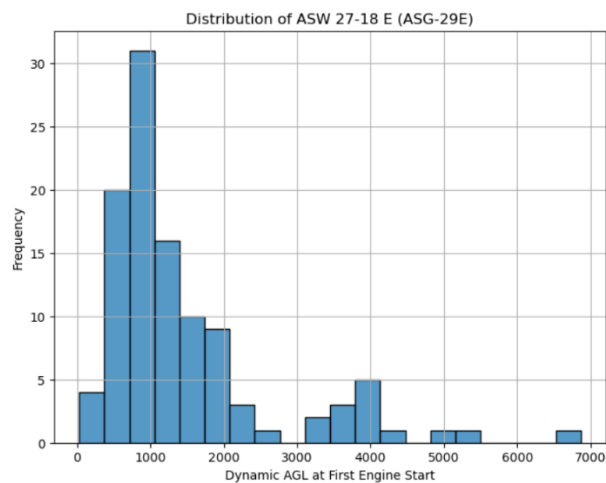
```
Min_%%_off  13.21
Max_%%_off  12.96
Avg_%%_off  11.00
```

We also completed analysis to understand which pilots are flying to standard (i.e. starting their engine above 1000 ft), and which pilots are not. To isolate pilot behavior, we are looking at the glider type, which should be unique to a pilot.

In the below plot, we are looking at the number of events where a pilot started their engine below a 1000 ft. Below are the top 10 gliders that have started their engine below standard the most. With the data available, ASW 27-18E (ASG-29E) had about 60 events of engine starts below 1000 ft.



However, we also built analysis to help us understand the distribution of the AGL at engine start per glider. In other words, ASW 27-18E (ASG-29E) had the highest number of engine-start events below 1000 ft. But how is their performance relative to all the engine-starts they encounter? Below histogram is looking at all the engine-starts that this glider type has experienced, with the data available. This shows that they do peak at 1000 ft and have a skewed distribution. To summarize for this glider, although they do tend to follow standard at times, they skew towards starting their engine too early (below a 1000 ft).



Future Work Plan

Task Description	Date
Compliance Program that outputs whether pilots are compliant	April 10
Progress Report 2	April 11
Rough Draft Presentation	April 24
Final Paper	April 30
Final Presentation	May 1

Potential Concerns [C] and Blockers [B]

Identifier	Description
False Positives	A false positive in our data is an instance in which a flight started its engine below 1000ft, but while the data showed that the engine ran, it wasn't a true engine run. We are unsure of exactly how to determine whether a false positive occurred other than filtering out Pure Gliders.
Successful Engine Run	We are unclear on how to truly determine if an engine start was successful beyond assessing if avg speeds were within the expected range and if altitude rate of change was not a large negative number.
Test engine runs	Pilots are instructed to test that their engine runs at the start of their flight. Our feedback is that it could be 4 to 5 minutes into the flight and will last about 1 minute of engine run time. From analysis completed so far, we are concerned that we can accurately capture if it was a "test" event based on data available.